

The Frattini Subgroup of Locally Free Groups

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1. Introduction

A group L is locally free (of finite rank m) if every finitely generated subgroup is contained in a free group (of rank m). We exclude the case $m=1$ for then L would be a subgroup of the rational numbers. There seems to be little known about locally free groups beyond what is given in [1, 3] and [5]; they are not simple, their commutator quotient group is torsion free and conversely, any torsion free abelian group is the commutator quotient of a locally free group, and, if of finite rank, they must be countable. This paper illustrates a technique that can sometimes be used to prove that the Frattini subgroup of locally free groups is trivial.

Since any locally free group of finite rank m is countable (see [5]), if L has rank m , then it must be an ascending union of free groups L_i of rank m , $i=1, 2, \dots$. Take as free generators of L_i , $\{x_{i1}, x_{i2}, \dots, x_{im}\}$. Since $L_{i-1} \subset L_i$, there are words W_{ik} , such that $x_{i-1k} = W_{i-1k}(x_{i1}, x_{i2}, \dots, x_{im})$, $k=1, 2, \dots, m$ and $i=1, 2, \dots$. We will illustrate our technique by proving a theorem for certain locally free groups of rank 2, and then state a result for locally free groups of arbitrary rank, which may be obtained by generalizing our method. So we assume $m=2$ (i.e. L has rank 2) and take $\{x_i, y_i\}$ to be free generators of L_i . If $W_{ik} = W_{i+1k}$ for $k=1, 2$ and $i=1, 2, \dots$, then we say L has constant embedding and replace W_{i1} and W_{i2} by W and V respectively. Therefore, if L is free of rank 2 and has constant embedding (as is assumed for the remainder of this section) then:

$$L = \bigcup_1^\infty L_i \text{ where each } L_i \text{ has free generators } x_i \text{ and } y_i \text{ and } x_i = W(x_{i+1}, y_{i+1}), \\ y_i = V(x_{i+1}, y_{i+1}).$$

It will sometimes be convenient when discussing the abstract words W and V to write them as words in symbols A and B , i.e., $W(A, B)$ and $V(A, B)$. For example if

$$x_i = W(x_{i+1}, y_{i+1}) = x_{i+1} y_{i+1} \quad \text{and} \quad y_i = V(x_{i+1}, y_{i+1}) = y_{i+1} x_{i+1},$$

then $W(A, B) = AB$ and $V(A, B) = BA$. We will without loss of generality assume that $\{W(A, B), V(A, B)\}$ is Nielsen reduced (N.R.) in A and B . Familiarity with [4], pp. 120–125, is assumed in this paper. We are particularly concerned with the form of the initial and terminal segments of the (freely reduced) words $W(A, B)$ and $V(A, B)$. We introduce the notation

$$W(A, B) = X(A, B) - Z(A, B)$$

to mean that $W(A, B)$ equals

$$X(A, B) Y(A, B) Z(A, B)$$

where $X(A, B)$, $Y(A, B)$, and $Z(A, B)$ are freely reduced words in A and B , and no cancellation occurs between $X(A, B)$ and $Y(A, B)$ or between $Y(A, B)$ and $Z(A, B)$. Equivalently, $X(A, B)$ and $Z(A, B)$ are disjoint initial and terminal segments respectively of $W(A, B)$. For example, if

$$W(A, B) = A^{-2} B^2 A B A^2 B^{-1}$$

then letting $X(A, B) = A^{-1}$ and $Z(A, B) = AB^{-1}$,

$$W(A, B) = A^{-1} - AB^{-1}.$$

Letting $X(A, B) = A^{-2} B$ and $Z(A, B) = BA^2 B^{-1}$ we also have

$$W(A, B) = A^{-2} B - BA^2 B^{-1}.$$

But it is not valid to choose $X(A, B) = A^{-2} B^2 A$ and $Z(A, B) = ABA^2 B^{-1}$ because X and Z are not disjoint.

If d is a member of L then for some positive integer k , d is a member of L_j for $j \geq k$. So for each such j , d has a unique representation, which denote by $(d)_j$, as a freely reduced word in the generators of L_j , x_j and y_j . For any pair $c, d \in L$ there is a smallest k such that $c, d \in L_k$. If for $j \geq k$ there is no cancellation between the terminal segment of $(c)_j$ and the initial segment of $(d)_j$ then we say that c is blocked off from d . Being blocked off is not symmetric.

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2. A Theorem for Rank 2

Theorem 1. *Let L be a locally free group of rank 2 with constant embedding given by words W and V . If $W(A, B)$ and $V(A, B)$ are Nielsen reduced words of the form*

$$W(A, B) = A^{-1} - B \quad \text{and} \quad V(A, B) = A^{-1} - AB^{-1},$$

then ΦL , the Frattini subgroup of L is trivial.

Proof. In order to prove that for any $z \in L$ there is a maximal subgroup excluding z , it suffices by Zorn's Lemma to exhibit a subgroup H of L excluding z , which together with z generates L . We do this first for $z = y_1$. If S is any subset of L , $\langle S \rangle$ will denote the subgroup generated by S . We define H_n to be the subgroup generated by the following elements:

$$\begin{array}{cc} y_1 x_2^{-1} y_1, & x_2 y_2 x_2^{-1} \\ y_2 x_3^{-1} y_2, & x_3 y_3 x_3^{-1} \\ & \vdots \\ y_{n-1} x_n^{-1} y_{n-1}, & x_n y_n x_n^{-1}. \end{array}$$

It is immediate from the definition that:

$$H_n \subseteq L_n, \quad (1)$$

$$H_n \subset H_{n+1}, \quad (2)$$

$$\langle H_n, y_1 \rangle = L_n, \quad (3)$$

$$\left\langle \bigcup_2^\infty H_n, y_1 \right\rangle = L. \quad (4)$$

We will show through several lemmas that H_n is free (of rank $2n-2$) on its given generating set from which it follows that H_n does not equal L_n which has rank 2. Then let $H = \bigcup_2^\infty H_n$. If $H = L$, then for some n , $y_1 \in H_n$, which is impossible. Therefore H has the required properties vis-à-vis y_1 . To prove that H_n is free on its generating set it suffices to show that the generating set, when expressed as words in x_n and y_n , is N.R. So we must show that there will not be too much cancellation among the generators.

We begin by examining H_2 . By definition,

$$H_2 = \langle y_1 x_2^{-1} y_1, x_2 y_2 x_2^{-1} \rangle.$$

Using $y_1 = V(x_2, y_2) = x_2^{-1} - x_2 y_2^{-1}$ and $x_1 = W(x_2, y_2) = x_2^{-1} - y_2$, the generators of H_2 may be rewritten in terms of x_2 and y_2 to give:

$$H_2 = \langle (x_2^{-1} - x_2 y_2^{-1}) x_2^{-1} (x_2^{-1} - x_2 y_2^{-1}), x_2 y_2 x_2^{-1} \rangle. \quad (5)$$

It is clear that the two generators in (5) form an N.R. set in L_2 , that is, with respect to x_2 and y_2 . So assume H_n is N.R. in L_n and consider H_{n+1} . It must be proved that:

Statement (a). The generators of H_n remain N.R. when rewritten in x_{n+1} and y_{n+1} .

Statement (b). Adding the (two) new generators to H_n still gives a N.R. set.

Although it is not in general true that when the letters of an N.R. set of words are replaced by N.R. words, the resulting set of words is N.R., it is true in the special case we must consider. The following lemma will be used to verify Statement (a).

Lemma 1. Every generator of H_n is of the form $cdc^{\pm 1}$ where $c, d \in L$, and c is blocked off from d , and d is blocked off from $c^{\pm 1}$.

Proof. For a generator of the form $x_j y_j x_j^{-1}$ let $c = x_j$ and $d = y_j$. Now write on successive lines $(c)_j \cdot (d)_j$, $(c)_{j+1} \cdot (d)_{j+1}$, $(c)_{j+2} \cdot (d)_{j+2}$, etc. (see introduction for notation and terminology). At each step there is a letter of $(c)_k$ adjacent to a letter of $(d)_k$. The successive adjacent letters of $(c)_k$ and $(d)_k$ respectively for $c = x_j$ and $d = y_j$, $j \leq k \leq j+3$, are:

$$\begin{aligned} & x_j \cdot y_j, \\ & y_{j+1} \cdot x_{j+1}^{-1}, \\ & y_{j+2}^{-1} \cdot y_{j+2}^{-1}, \\ & x_{j+3} \cdot y_{j+3}. \end{aligned}$$

Since L has constant embedding, the adjacent letters $x \cdot y$ (disregard subscripts) will repeat periodically without cancellation occurring between $(c)_k$ and $(d)_k$. This argument works in all cases. Note that cancellation within a term $(c)_k$ will not interfere with this argument because W and V are N.R.

We can now prove that Statement (a) holds. All of the generators of H_n are of the form $cdc^{\pm 1}$, for various c 's and d 's where by Lemma 1, the c 's and d 's do not interact with each other when rewritten in x_{n+1} and y_{n+1} . Of course any cancellation, in a particular generator, occurring within the left c also occurs within the right one. Now suppose two generators of H_n expressed in x_{n+1} and y_{n+1} share an initial segment of length t . This segment cannot include an entire c since W and V are N.R. Therefore the length of either generator must exceed $2t+1$, from which it easily follows that the generators of H_n remain N.R.

We now turn our attention to Statement (b). For $n > 1$ let T_{n+1} be the set of generators of H_{n+1} which are also generators of H_n and let R_{n+1} be the set of new generators of H_{n+1} . Since R_{n+1} , with its members rewritten in terms of x_{n+1} and y_{n+1} , is clearly N.R., we have only to verify that any element g in either T_{n+1} or R_{n+1} satisfies the following conditions (see [4], p. 123):

C_1 . The major initial and terminal segments of $(g)_{n+1}$ are isolated from the elements in the other set.

C_2 . If $(g)_{n+1}$ has even length, then its right and left halves are isolated from the elements of the other set.

In C_1 and C_2 it is assumed that all elements are written as freely reduced words in x_{n+1} and y_{n+1} .

We first consider the case $g \in R_{n+1}$, so g is either $y_n x_{n+1}^{-1} y_n$ or $x_{n+1} y_{n+1} x_{n+1}^{-1}$. Therefore $(g)_{n+1}$ is either

$$(x_{n+1}^{-1} - x_{n+1} y_{n+1}^{-1}) x_{n+1}^{-1} (x_{n+1}^{-1} - x_{n+1} y_{n+1}^{-1}) \quad \text{or} \quad x_{n+1} y_{n+1} x_{n+1}^{-1}.$$

Since both have odd length, it must only be shown that C_1 holds. This is immediate for $x_{n+1} y_{n+1} x_{n+1}^{-1}$; since $(W(A, B))^{\pm 1}$ and $(V(A, B))^{\pm 1}$ do not begin with A or end with A^{-1} , $x_{n+1} y_{n+1}^{\pm 1}$ can never appear as an initial segment and $y_{n+1}^{\pm 1} x_{n+1}^{-1}$ can never appear as a terminal segment. Lemmas 2 and 3 show that major initial and terminal segments of $(x_{n+1}^{-1} - x_{n+1} y_{n+1}^{-1}) x_{n+1}^{-1} (x_{n+1}^{-1} - x_{n+1} y_{n+1}^{-1})$ are also isolated. This is a consequence of the form of $W(A, B)$ and $V(A, B)$ and the fact that they are N.R. It does not depend on T_{n+1} .

Lemma 2. *A word of the form $(y_{n+1} x_{n+1}^{-1} - x_{n+1}) x_{n+1}$ does not occur as an initial segment in $\langle T_{n+1} \rangle$.*

Proof. Since $(y_{n+1}^{-1})_{n+1} = y_{n+1} x_{n+1}^{-1} - x_{n+1}$, if $(y_{n+1} x_{n+1}^{-1} - x_{n+1}) x_{n+1}$ occurs as an initial segment, there is an element $v \in L_n$ such that $(y_{n+1}^{-1} v)_{n+1}$ has $(y_{n+1} x_{n+1}^{-1} - x_{n+1}) x_{n+1}$ as initial segment. But then $(v)_{n+1}$ is a word beginning with x_{n+1} . There can be no such v since $(W(A, B))^{\pm 1}$ and $(V(A, B))^{\pm 1}$ never begin with A .

Lemma 3. *A word of the form $(x_{n+1}^{-1} - x_{n+1} y_{n+1}^{-1}) x_{n+1}^{-1}$ does not occur as an initial segment in $\langle T_{n+1} \rangle$.*

Proof. Since $(y_n)_{n+1} = x_{n+1}^{-1} - x_{n+1} y_{n+1}^{-1}$, if $(x_{n+1}^{-1} - x_{n+1} y_{n+1}^{-1}) x_{n+1}^{-1}$ occurs as an initial segment, there is an element $v \in L_n$ such that $(y_n v)_{n+1}$ has $(x_{n+1}^{-1} - x_{n+1} y_{n+1}^{-1}) x_{n+1}^{-1}$

as initial segment. Then the word $(v)_{n+1}$ begins with x_{n+1}^{-1} , so that $(v)_n$ must begin with y_n or x_n and therefore either y_n^2 or $y_n x_n$ is an initial segment of $(y_n v)_n$. There can be no such v since $(W(A, B))^{\pm 1}$ and $(V(A, B))^{\pm 1}$ never begin with B^2 or BA .

Similar arguments account for $x_{n+1}^{-1}(x_{n+1}^{-1} - x_{n+1} y_{n+1}^{-1})$ and $x_{n+1}(y_{n+1} x_{n+1}^{-1} - x_{n+1})$ as terminal segments. This disposes of the case $g \in R_{n-1}$. We state this formally as:

Lemma 4. C_1 and C_2 are satisfied for $g \in R_{n+1}$.

We now have:

Lemma 5. C_1 and C_2 are satisfied for $g \in T_{n+1}$.

Proof. The essence of the argument is that for $r \in R_{n+1}$, $l_{n+1}(g)$, the length of $(g)_{n+1}$ in the generators x_{n+1} and y_{n+1} , is too large relative to $l_{n+1}(r)$ for $(r)_{n+1}$ to have a large initial or terminal segment of $(g)_{n+1}$ as an initial or terminal segment. Because the embedding $L_i \subset L_{i+1}$ is the same for all i , it will suffice to show that C_1 and C_2 hold for $g \in T_3$ (that is $g \in H_2$) and $r \in R_n$, $n \geq 3$. Let l_w and l_v be the lengths of $W(A, B)$ and $V(A, B)$ respectively in A and B . Since for $n=3$, r equals either $y_2 x_3^{-1} y_2$ or $x_3 y_3 x_3^{-1}$, it follows that $l_3(r)$ equals $2l_v + 1$ or 3 . By assumption $g \in \{y_1 x_2^{-1} y_1, x_2 y_2 x_2^{-1}\}$. Suppose $g = y_1 x_2^{-1} y_1$. Then $l_3(g) > 2l_v + 2$. Therefore if $(g)_3$ and $(r)_3$ shared a half of $(g)_3$, and a fortiori a major segment of $(g)_3$, then Lemma 4 would be violated. Since for $n > 2$, $l_{n+1}(g) \geq l_3(g)$, Lemma 5 is therefore proved for $g = y_1 x_2^{-1} y_1$.

For $g = x_2 y_2 x_2^{-1}$ we have $l_4(g) > 3l_v$, so the previous argument may be employed once it is verified that $(g)_3$ is properly isolated from R_3 . This is easily done, and the lemma follows.

Lemmas 4 and 5 imply Statement (b), which together with Statement (a) imply that H_n is free (of rank $2n-2$) on its given set of generators. Therefore $H = \bigcup_2^\infty H_n$ has the required properties. This completes the proof of Theorem 1 for $z = y_1$.

Now suppose z is an arbitrary element of L . Since L is an ascending union, there is no loss of generality in assuming that $z \in L_2$. We alter H_n by eliminating $y_1 x_2^{-1} y_1$ and replacing x_2 with $x_2^{-k} z x_2^k$ (the one place x_2 occurs in the definition of H_2 after $y_1 x_2^{-1} y_1^{-1}$ has been eliminated), where k is a positive integer sufficiently large so that

$$(x_2^{-k} z x_2^k)_2 = x_2^{-1} - x_2.$$

Otherwise H_n is defined exactly as before. Then an almost identical proof, with $x_2^{-k} z x_2^k$ assuming the role of y_1 , shows that H which is defined as before, is a proper subgroup of L , while $\langle H, x_2^{-k} z x_2^k \rangle = L$. A sketch of this proof follows.

The element $x_2 y_2 x_2^{-1}$ is replaced by

$$(x_2^{-k} z x_2^k) y_2 (x_2^{-k} z x_2^k)^{-1} = x_2^{-k} z x_2^k y_2 x_2^{-k} z^{-1} x_2^k = x_2^{-1} - x_2 y_2 x_2^{-1} - x_2.$$

Since x_2 is blocked off from y_2 and y_2 from x_2^{-1} , Lemma 1 and therefore the validity of Statement (a) follow as before. As observed earlier Lemmas 2, 3, and therefore Lemma 4 depend only on $\{W(A, B), V(A, B)\}$; the earlier proofs are valid for the present case exactly as they stand. Finally to prove Lemma 5, we need only adapt the original proof to the case $g = (x_2^{-k} z x_2^k) y_2 (x_2^{-k} z x_2^k)^{-1}$. We conclude that H_n is free and not equal to L_n . Obviously $\langle H_n, x_2^{-k} z x_2^k \rangle = L_n$, so letting $H = \bigcup_2^\infty H_n$, it follows that $\langle H, x_2^{-k} z x_2^k \rangle = L$, while $H \neq L$. This implies the existence of a maxi-

mal subgroup of L excluding $x_2^{-k} z x_2^k$, from which it follows that the same is true for z . Therefore $\Phi L = 1$. The proof of Theorem 1 is complete.

Corollary 1. *The locally free group of rank 2 determined by $W(A, B) = A^{-1} B^{-1} A B$ and $V(A, B) = A^{-1} B A B^{-1}$ has trivial Frattini subgroup.*

Observe that the group of Corollary 1 is perfect.

3. Arbitrary Rank

The method employed to prove Theorem 1 may be extended to prove the triviality of the Frattini subgroup of certain locally free groups of arbitrary finite rank, even when the embedding is not constant. We state one such result without proof. First we must define the simple n -fold commutator $[x_1, x_2, \dots, x_n]$:

$$[x_1, x_2] = x_1^{-1} x_2^{-1} x_1 x_2,$$

$$[x_1, x_2, \dots, x_n] = [[x_1, \dots, x_{n-1}], x_n].$$

Theorem 2. *If L is the locally free group of rank $m > 3$ given by:*

$$W_{i-1j}(x_{i1}, x_{i2}, \dots, x_{im}) = [x_{ij}, x_{ij+1}, \dots, x_{im}, x_{i1}, x_{i2}, \dots, x_{ij-1}]^i,$$

then $\Phi L = 1$.

While other results like Theorems 1 and 2 may be stated and proved similarly, there are locally free groups of finite rank for which the embedding $L_i \subset L_{i+1}$, seem to preclude our approach. It may be that the triviality of the Frattini subgroup of locally free groups follows from more general considerations than discussed here.

References

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